Meridian Supply Chain Intelligence RAG System

HCLTech Assignment 2 — Comprehensive Final Technical & Audit Report

Author / Candidate:	Esambadi Goutham Reddy	**Target Entity:**	Meridian Components Pvt. Ltd.
GitHub Repository:	goutham-11-16/Supply-Chain-RAG	**LLM & Embeddings:**	GPT-4o & text-embedding-3-small
Vector Store:	ChromaDB (supplychain_rag)	**Indexed Vectors:**	22 Chunks (Size: 1200, Overlap: 150)

1. Project Overview

This project establishes a production-quality, document-grounded Retrieval-Augmented Generation (RAG) system for **Meridian Components Pvt. Ltd.**, an automotive electronics manufacturer operating assembly plants in Chakan (Pune) and Hosur (Tamil Nadu). Meridian's procurement team handles complex buyer queries regarding supplier performance metrics (on-time delivery, defect PPM, line stoppages) and corporate policy rules (PO approval tiers, quality cost recovery under Clause 6.3, and safety stock formulas).

Key Problem Addressed: Manual inspection of fragmented PDF handbooks leads to delayed decision-making, misinterpretation of quality penalties (e.g., failing to enforce mandatory 100% inspection floors), and high risk of AI hallucinations when using ungrounded LLMs.

Core RAG Capabilities: The system indexes all Meridian PDFs into a persistent single-collection ChromaDB vector database, supports configurable Top-K similarity search (1–12), provides dual-column source page citations, integrates a deterministic Policy & Penalty Calculator, and strictly refuses out-of-domain trap questions.

2. System Architecture

The application architecture enforces strict separation between data ingestion, vector indexing, similarity retrieval, and LLM context synthesis:

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PDF Documents in data/
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▼
PyPDF Extraction (0-indexed metadata standardized to 1-indexed pages)
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▼
Recursive Character Text Splitter (Chunk Size: 1200, Overlap: 150)
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▼
OpenAI text-embedding-3-small (1536-dimensional vectors)
■
▼
ChromaDB Persistent Vector Store (Single Collection: supplychain_rag)
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▼
Top-K Similarity Retrieval (Configurable 1–12, Default: 6 with Smart Balancing)
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▼
Strict System Prompt + Context Construction
■
▼
OpenAI GPT-4o Synthesis (Temperature = 0.1)
■
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▼
Grounded Answer + Dual-Column Source Citations Audit Trail
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3. Document & Dataset Details

Document Name	Document Type	Pages	Key Content / Information Covered
Meridian_Procurement_Policy_Handbook_v4.2.pdf	Corporate Policy	3 Pages	Approval authority tiers, supplier classification, Clause 6.1 OTD warnings, Clause 6.3 quality penalties (■120/unit), Section 8 safety stock formulas.
Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf	Operational Review	3 Pages	Q1 scorecard metrics (OTD, PPM) for 5 key suppliers, 7 line stoppage events (41 hrs total), single-source microcontroller risks, freight lanes.

Chunking Hyperparameters: Chunk Size = 1200 characters, Chunk Overlap = 150 characters, Total Chunks = 22. *Rationale: Preserves complete scorecard data tables and policy clauses in a single vector chunk without splitting numerical context.*

4. Retrieval Strategy

The retrieval pipeline queries ChromaDB using `text-embedding-3-small` similarity vectors. Top-K defaults to 6 (slider range 1–12). To prevent cross-document queries from being dominated by a single document, the engine applies a smart cross-document candidate merger and full-content string deduplication, ensuring balanced context representation from both PDFs.

5. Cross-Document Reasoning

Cross-document reasoning is a core requirement of the Meridian RAG system. The engine seamlessly combines operational facts from the Review PDF with governing clauses from the Handbook PDF:

- **Kaveri Metals:** Combines Review Q1 (88.1% OTD, 1,150 PPM) with Handbook Clause 6.1 (written warning & weekly review calls) and Clause 6.3 (■120/unit rework debit + mandatory 100% inspection floor until 3 consecutive lots pass).
- **Trident Circuit Boards:** Combines Review Q1 (640 PPM defect rate, 2 line stoppages) with Handbook Clause 6.3 penalty consequences.
- **Shenzhen Rui Electronics:** Combines Review Q1 (Highest spend ■21.9 cr, 76.0% OTD, single-source microcontroller risk) with Handbook Clause 7.1 (mandatory 12-month dual-sourcing rule) and current company actions (Anh Long Vietnam qualification).

6. Policy & Penalty Calculator

Operating alongside the RAG pipeline, Tab 3 of the application features a deterministic execution engine that computes exact policy actions and penalties:

- **Clause 6.1 (OTD Warning):** Triggers written warning & weekly review calls if OTD < 90%.
- **Clause 6.2 (Rating Bands):** Band A (≥90%), Band B (75–89%), Band C (60–74%), Band D (<60%).
- **Clause 6.3 (Quality Recovery):** PPM > 500 triggers ■120/unit rework recovery debit and 100% incoming inspection floor.
- **Section 8 (Safety Stock):** Computes *SS = max(Lead Time × 0.25, Floor Days)*. For imported critical parts (46-day LT), *max(11.5, 30) = 30 days*.

7. Comprehensive 20-Test Benchmark Suite

All 20 test cases were executed against the live system. Every query returned 100% accurate, document-grounded results:

ID	Category	User Question	Expected Output Summary	Status
T1	Single-Doc	Highest spend supplier in Q1 & OTD?	Shenzhen Rui Electronics (■21.9 cr, 76.0% OTD)	**■ PASS**
T2	Single-Doc	Line stoppages, downtime & causes?	7 stoppages, 41 hrs downtime, 4 micro & 2 PCB causes	**■ PASS**
T3	Single-Doc	PO approval authority for ■1.4 crore?	Chief Operating Officer (COO) (Tier ■1cr–■5cr)	**■ PASS**
T4	Single-Doc	4 classification categories & Critical criteria?	Strategic, Critical, Standard, Tactical; single-source/LT>30d	**■ PASS**
T5	Cross-Doc	Kaveri Metals 88.1% OTD & 1,150 PPM clauses?	Clause 6.1 warning & Clause 6.3 ■120 debit + 100% insp floor	**■ PASS**
T6	Cross-Doc	Single-source microcontroller policy & actions?	Clause 7.1 12mo dual source; Anh Long Vietnam validation	**■ PASS**
T7	Cross-Doc	46-day lead time safety stock required?	max(11.5, 30) = 30 days safety stock	**■ PASS**
T8	Cross-Doc	Trident 640 PPM defect consequence?	Clause 6.3 ■120/unit rework & 100% inspection floor	**■ PASS**
T9	Cross-Doc	Suppliers below Band B OTD & escalation?	None (all ≥75%); Band C needs plan, Band D business hold	**■ PASS**
T10	Trap Test	Annual salary of Head of Procurement?	'Information is not available in the uploaded documents.'	**■■ PASS**
T11	Single-Doc	PO approval authority for ■25 lakh?	Head of Procurement (Tier ■10L–■50L)	**■ PASS**
T12	Single-Doc	Freight lane with longest transit time?	Shenzhen to Nhava Sheva (28 days ocean transit)	**■ PASS**
T13	Single-Doc	Apex Microelectronics Q1 OTD & PPM?	91.2% OTD, 210 PPM defect rate (Band A)	**■ PASS**
T14	Single-Doc	Share-of-wallet cap for single supplier?	Maximum 60% of total category spend	**■ PASS**
T15	Cross-Doc	Policy actions for Sunrise Logistics Q1 OTD?	94.0% OTD (Band A); no penalty triggered	**■ PASS**
T16	Cross-Doc	Emergency air freight approval authority?	Requires VP of Supply Chain written approval	**■ PASS**
T17	Trap Test	What is Meridian's Q3 revenue forecast?	'Information is not available in the uploaded documents.'	**■■ PASS**
T18	Trap Test	Who is the CEO of Meridian Components?	'Information is not available in the uploaded documents.'	**■■ PASS**
T19	Single-Doc	How many days customs hold at Nhava Sheva?	9 days customs hold (11 hours line stoppage)	**■ PASS**
T20	Cross-Doc	What recovery rate applies to defective parts?	Standard recovery rate of ■120 per affected unit	**■ PASS**

Application Interface Screenshots

8. Retrieval / Top-K Validation

To verify that the retrieval pipeline strictly respects the user-selected Top-K parameter without arbitrary trimming or leakage, we executed a systematic empirical validation across all supported Top-K values (1 to 12):

Top-K Setting	Raw Chroma Chunks	Filtered Chunks	Deduped Chunks	LLM Context Chunks	Debug UI Displayed	Match Status
Top-K = 1	1	1	1	1	1	**■ 1-to-1 Exact**
Top-K = 2	2	2	2	2	2	**■ 1-to-1 Exact**
Top-K = 4	4	4	4	4	4	**■ 1-to-1 Exact**
Top-K = 6 (Default)	6	6	6	6	6	**■ 1-to-1 Exact**
Top-K = 8	8	8	8	8	8	**■ 1-to-1 Exact**
Top-K = 10	10	10	10	10	10	**■ 1-to-1 Exact**
Top-K = 12	12	12	12	12	12	**■ 1-to-1 Exact**

9. Hallucination / Trap Test

Test Query: *'What is the annual salary of the Head of Procurement?'*
System Output: **'The information is not available in the uploaded documents.'**
Sources: None (0 chunks referenced).
Significance: Demonstrates that the system enforces strict prompt boundaries and refuses out-of-domain questions rather than inventing figures.

10. Development History & Known Limitations

• **Deduplication Header Fix:** Early testing revealed that deduplication using string slicing (`doc[:100]`) compared prepended metadata headers rather than actual body text. This was corrected to compare `doc.page_content.strip()`, restoring 1-to-1 Top-K counts.
• **Windows File Locking Fix:** Replaced file-system folder deletion (`shutil.rmtree`) with Chroma's native `existing.delete_collection()` API in `ingest.py` to prevent vector duplication caused by Windows file locks.
• **Domain Scope:** System is tailored to Meridian's supply chain documents and strictly refuses un-indexed queries.

11. Final Verification Checklist

Verification Item	Requirement	Status
Single Collection Persistence	Both Meridian PDFs stored in 1 ChromaDB collection	**■ VERIFIED**
Chunking Count Accuracy	22 unique vector chunks generated & reported	**■ VERIFIED**
Top-K Range Control	Exact 1-to-1 retrieval for K = 1, 2, 4, 6, 8, 10, 12	**■ VERIFIED**
Cross-Document Reasoning	Synthesizes Review Q1 metrics + Handbook policy rules	**■ VERIFIED**
Source Citations	Provides document name & 1-indexed page for every fact	**■ VERIFIED**
Policy & Penalty Calculator	Evaluates OTD bands, ■120 rework debit & safety stock math	**■ VERIFIED**
Trap Question Refusal	Honest refusal notice for un-indexed queries	**■ VERIFIED**
API Key Security	.env listed in .gitignore; zero exposed secrets on GitHub	**■ VERIFIED**

12. Conclusion

The Meridian Supply Chain Intelligence RAG System successfully satisfies all technical, architectural, functional, and security requirements of HCLTech Assignment 2. Built by **Esambadi Goutham Reddy**, the system combines robust vector retrieval, grounded GPT-4o synthesis, deterministic policy evaluation, and strict security compliance into an executive-ready application.

Report generated autonomously for HCLTech Assignment 2 Submission Verification.